

3.2 The Derivative as a Function Worksheet

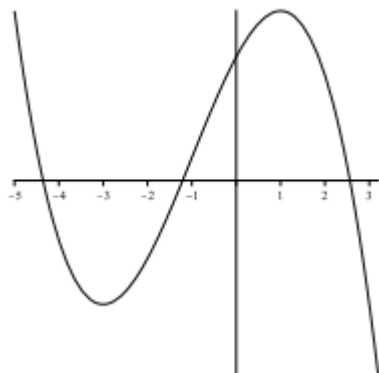
1. Find the derivative of the function using the definition of the derivative. State the domain of the function and the domain of its derivative.

a) $f(x) = \frac{1}{2}x - \frac{1}{3}$

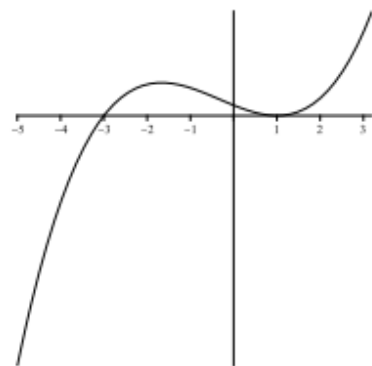
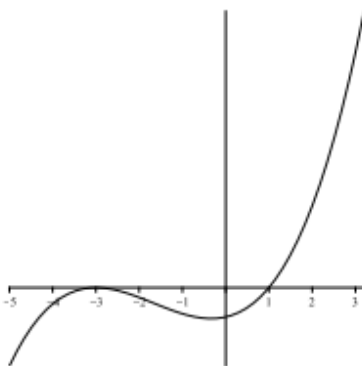
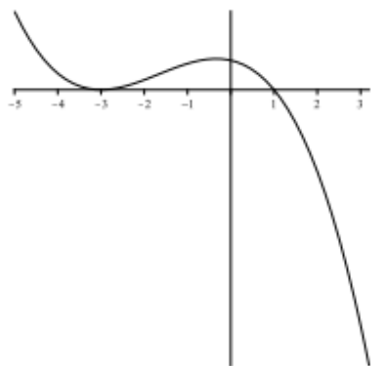
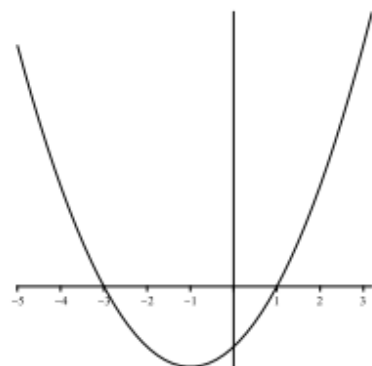
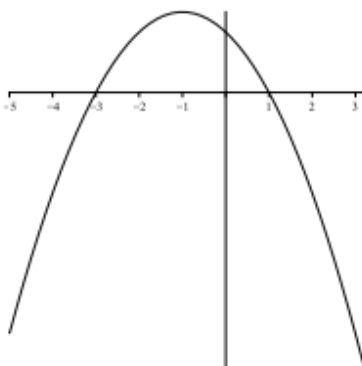
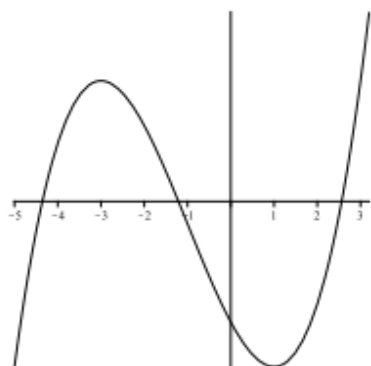
b) $f(x) = \sqrt{9-x}$

c) $f(x) = x^{3/2}$

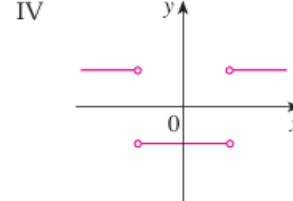
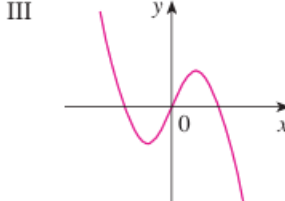
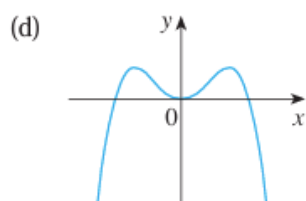
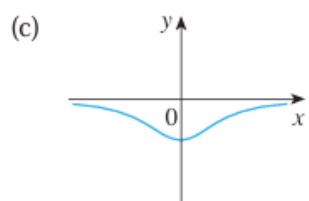
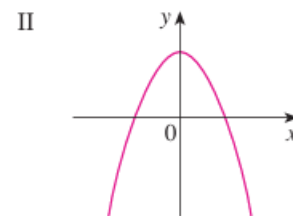
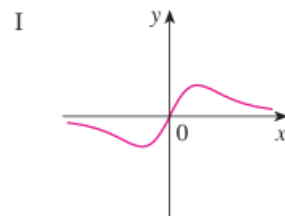
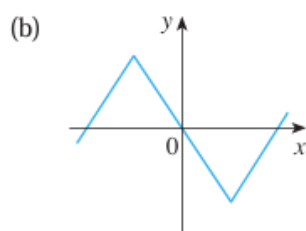
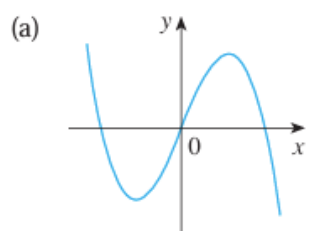
2. The graph of the function $f(x)$ is shown below.



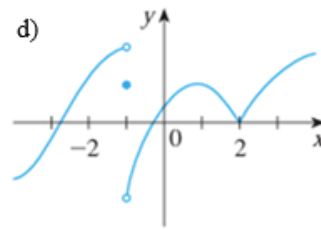
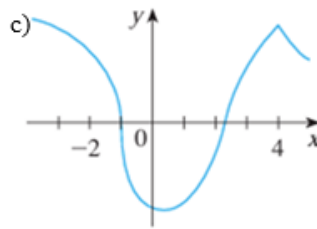
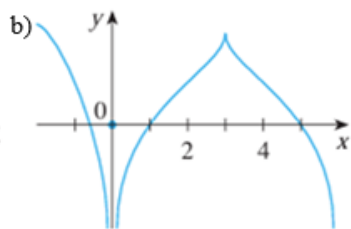
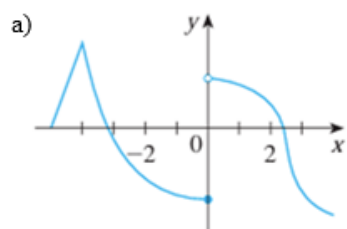
One of the six graphs below is the graph of $f'(x)$. Circle the graph of $f'(x)$.



3. Match the graph of each function in (a) – (d) with the graph of its derivative in I-IV.



4. The graph of f is given. State, with reasons, the numbers at which f is not differentiable.



5. Suppose that we know that $g(x)$ is an *even* function that is differentiable at $x = 1$. we know that $g(x)$ has the following values:

$$g(-4)=1 \quad g(-2)=1 \quad g(-1)=3 \quad g(0)=3 \quad g(2)=1 \quad g(3)=1$$

Find $\lim_{x \rightarrow 1} g(x)$.

6. TRUE or FALSE: If a function $f(x)$ is continuous at $x = c$, then $f'(x)$ must exist.